**Modern Academy**
for Engineering and Technology
Basic Science Department

**Academic Year:** 2017-2018
**Semester:** Fall
**Exam Date:** 15/1/2018

**QUESTIONS FOR THE FINAL WRITTEN EXAM**

**Subject:** (MTH203) Differential Equations and Transforms **Spec:** Comp. and Comm.
**Examiners:** Assoc. Prof. Dr. Ashraf Taha and Prof. Dr. Aly Essawi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

## Question 1 (14 points)

***Solve the following differential equations:***

a) $\cos(x)\cos(y)\dfrac{dy}{dx} = -\sin(x)\sin(y).$ (2 points)

b) $\dfrac{dy}{dx} = \dfrac{2xy}{3x^2 - y^2}.$ (4 points)

c) $y(x+y+1)dx + x(x+3y+2)dy = 0.$ (4 points)

d) $\dfrac{dy}{dx} - \dfrac{3}{x}y = \dfrac{1}{2x^3}y^3.$ (4 points)

## Question 2 (16 points)

***Find the general solution of the following differential equations:***

a) $\dfrac{d^3y}{dx^3} - 3\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} - y = 0.$ (4 points)

b) $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} - 2y = 2x - 40\cos(2x).$ (4 points)

c) $\dfrac{d^2y}{dx^2} + y = \sec(x).$ (4 points)

d) $\dfrac{d^2y}{dx^2} - \dfrac{1}{x}\dfrac{dy}{dx} + \dfrac{y}{x^2} = 3x^3 + 2,$ given that $y = x$ is a homogenous solution. (4 points)

## Question 3 (16 points)

a) ***Solve the following differential equation using Laplace transform:***

$\dfrac{d^2y}{dx^2} - 10\dfrac{dy}{dx} + 9y = 5; \qquad y(0) = -1, y'(0) = 2.$ (4 points)

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b) ***Find the Laplace transform of the following functions:***

i. $f(t)=\sinh 5t+\sin t\cos t.$ (3 points)

ii. $f(t)=\int_0^t 7\tau^3+2\cos 4\tau\ d\tau.$ (3 points)

iii. $f(t)=u_7(t)\ e^t\cos(t-7).$ (3 points)

iv. $f(t)=t^{11}\ e^{5t}+\dfrac{\cos 4t-\cos 5t}{t}.$ (3 points)

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**Question 4 (16 points)**

a) ***Find the inverse Laplace transform of the following functions:***

i. $F(s)=\ln\left(\dfrac{s+3}{s+1}\right)+\tan^{-1}\left(\dfrac{s}{2}\right).$ (4 points)

ii. $F(s)=\dfrac{3}{s^2+4s+13}.$ (4 points)

iii. $F(s)=e^{-2s}\left(\dfrac{2s+1}{s^2}\right).$ (4 points)

b) ***Represent the following function*** $f(x)=x$, $-\pi\le x\le\pi$ ***as Fourier Series*** (4 points)

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**Question 5 (8 points)**

a) ***Prove the recurrence relations*** $xP_n(x)=\dfrac{n+1}{2n+1}P_{n+1}(x)+\dfrac{n}{2n+1}P_{n-1}(x).$ (4 points)

b) ***Evaluate the integral*** $\int_{-1}^{1} P_3^2(x)\ dx.$ (2 points)

c) ***Evaluate the integral*** $\int_{-1}^{1} x\,P_1(x)P_2(x)\ dx.$ (2 points)

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***GOOD LUCK***